# Do Now · retrieval
## Character sets: ASCII & Unicode

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which data type should store a telephone number beginning with 0? Explain briefly.

### Q2

Convert $41_{16}$ to denary and eight-bit binary.

### Q3

Apply AND 00001111 to 10100101. Explain which bits the mask preserves.

### Q4

In 6+4 two’s-complement floating-point format, subtract 010000 0010 from 011000 0010. Give a normalised result.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Characters need agreed codes — A character set defines characters and their assigned numeric codes. An encoding specifies how those codes are represented as bytes.

## Core idea 2

Standard ASCII is limited — Seven bits provide $2^7 = 128$ codes for letters, digits, punctuation and control characters. — Eight-bit extensions can provide 256 values, but different extensions may map the extra codes differently.

## Core idea 3

Unicode supports a much larger repertoire — Unicode assigns code points to characters from many writing systems and symbols. Its first 128 code points match ASCII.

## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Core idea 4

Use the encoding stated in storage questions — Storage depends on the number of code points and the number of bytes used for each one. Do not assume every Unicode character uses the same width. — CODE:
'0' in ASCII = 48 = $00110000_2$
integer 0 = 0

## Vocabulary · use the correct technical terms

Character set — A collection of characters with assigned codes.

Code point — A numeric identifier assigned to a character in Unicode.

Encoding — Rules for representing character codes as bytes.

ASCII — A standard seven-bit character code with 128 possible values.

## Independent skill check · apply the lesson

Compare standard ASCII with Unicode for a multilingual app. Then encode the character C as an 8-bit ASCII byte (A = 65).

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Encode the text “AB”

## Follow the worked method

1. Start with the character codes — CODE:
A = 65
B = 66 — ASCII assigns a numeric code to each character.

2. Write the 7-bit ASCII values — CODE:
A = 1000001
B = 1000010

3. Add a leading zero — CODE:
A → 01000001
B → 01000010

AB → 01000001 01000010 — The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

## YOUR TASK

Build the answer

## CLUES · uncover one at a time

Clue 1: The ASCII code for character '0' is 48.

Clue 2: The code for character '7' is seven more than the code for '0'.

Clue 3: An integer with value 7 is not the same as the ASCII code for '7'.

## BUILD YOUR ANSWER · in BLACK PEN

Step 1: Find the ASCII code for the digit character.

Step 2: Convert that code to an 8-bit byte.

Step 3: Convert the integer 7 and explain the difference.

## CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Question 2 · ADAPTED EXAM COMPARISON · [4]
A school app must store English and Chinese messages. Explain why standard ASCII is unsuitable [1], why Unicode is suitable [1], give one similarity between them [1], and explain why the receiver must know the encoding [1].
Contextual adaptation of OCR June 2022 Q5(e)(ii–iii); classroom marks.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Question 5 · ORIGINAL ENCODING APPLICATION · [4]
A UTF-8 string contains 20 ASCII code points, three euro-sign code points and two emoji code points. The byte widths are 1, 3 and 4 respectively. Ignore metadata. Calculate bytes [2] and bits [1]. Explain why multiplying the number of code points by 2 bytes is not valid here. [1]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Standard ASCII provides 128 character codes.

## Use key vocabulary from memory

1. Explain Character set in your own words.

2. Explain Code point in your own words.

3. Explain Encoding in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

## END SELF-CHECK · only after you have attempted the work

**PAGE 5 · Guided We Do solution:**

Character '7' has code 55 = 00110111; integer 7 is 00000111 as an 8-bit unsigned value.

**PAGE 3 · Independent skill check:**

Standard ASCII has 128 codes (7-bit), unsuitable for Chinese characters; Unicode has a far wider repertoire. C = 67 = 01000011 in an 8-bit byte.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-008/answers.html